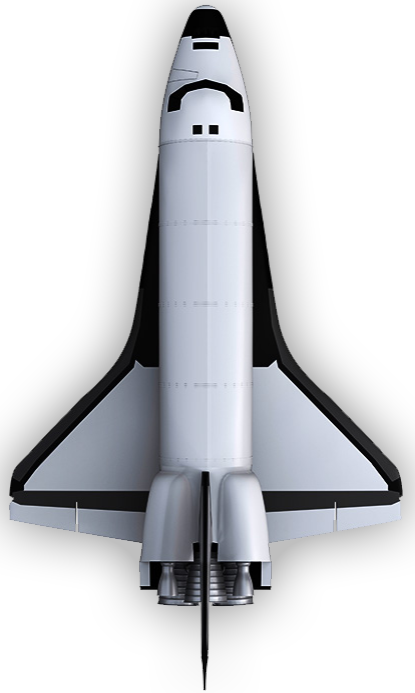
EEEE4064
ADVANCED CONTROL SYSTEM DESIGN

Lecture 8
Controller Design Using Pole Placement - 2





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Lecture Outline

01 Design of Servo Systems

02 Design of Type 1 Servo Systems

03 Type 1 Servo Systems – Case A

04 Type 1 Servo Systems – Case B

Design of Servo Systems

CONCEPT

- In the previous lecture, we have discussed the design of regulators with a constant input (zero). In this lecture, we will discuss the design of servo systems or servomechanisms, which are capable of closely following a changing input.
- Servo systems are feedback systems that generally have adjustable gains to achieve the desired output performance.

SERVO TYPES

- A **Type 0** servo has an error at rest. However, most servo applications require zero error at steady-state, so a Type 0 servo is generally unsatisfactory.
- A **Type 1** servo has zero error at steady-state due to the use of an integrator (we will focus on this type).
- A **Type 2** servo offers less errors with its two integrators, but causes a 180-degree lag around the full loop and increases the complexity of the implementation.

Design of Type 1 Servo Systems

CASE A

This is the case when the plant includes an integrator. In this case, we do not need to include an integrator in the control loop.

CASE B

If the plant has no integrator (type 0 plant), the basic principle of the design of a Type 1 servo system is to insert an integrator in the feedforward path between the error comparator and the plant.

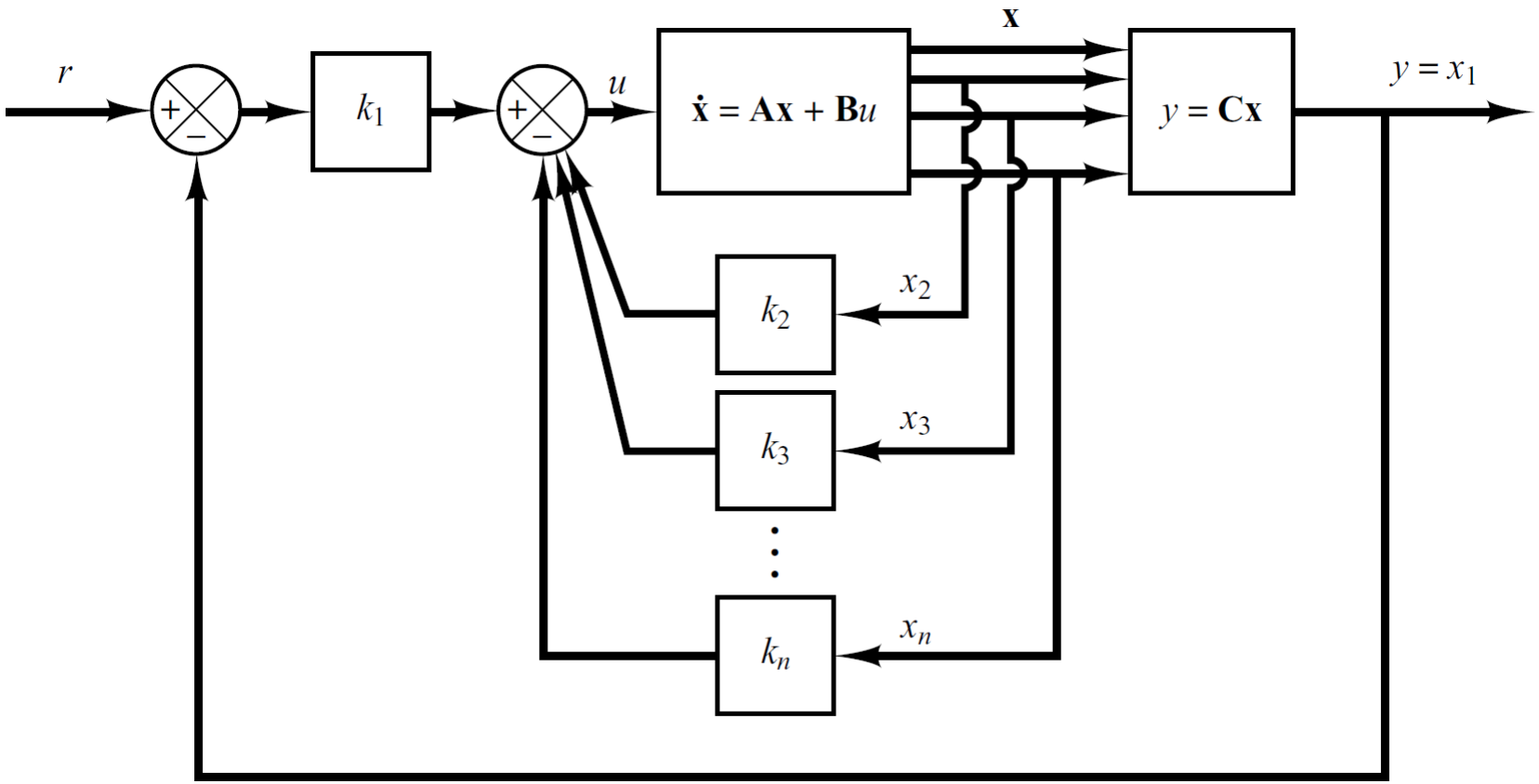
RULES

- We assume that both the control signal u and the output signal y are scalars, and the system is completely state controllable.
- The design process can be carried out using the same methods that we have discussed previously, i.e., the transformation matrix, direct substitution, and Ackermann's formula.

Type 1 Servo Systems – Case A

STRUCTURE

A Type 1 – Case A servo system when the plant has an integrator can be represented as follows:



Type 1 Servo Systems – Case A

RELATIONSHIPS

Consider the completely state controllable system:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

The control signal u:

$$u = -\begin{bmatrix} 0 & k_2 & k_3 & \cdots & k_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + k_1 \left(r - x_1 \right) = -\mathbf{K}\mathbf{x} + k_1 r$$

where:

$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & \cdots & k_n \end{bmatrix}$$

The system dynamics can be represented as follows:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x} + \mathbf{B}k_1 r$$

Type 1 Servo Systems – Case A

RELATIONSHIPS

The error dynamics will be:

$$\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{BK})\mathbf{e}$$

The steady-state value of the control signal:

$$u(\infty) = -\mathbf{K}\mathbf{x}(\infty) + k_1 r = 0$$

Type 1 Servo Systems – Case A

EXAMPLE 1

Design a type 1 servo system for the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Assume that the desired closed-loop poles are at:

$$s = -2 + j2\sqrt{3}, \quad s = -2 - j2\sqrt{3}, \quad s = -10$$

Then, investigate the response of the system using MATLAB using a step input.

Type 1 Servo Systems – Case A

Find the poles of the system:

$$|s\mathbf{I}-\mathbf{A}|=\begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 2 & s+3 \end{vmatrix}=s^3+3s^2+2s+1=s(s+1)(s+2)=0$$

The system includes an integrator. Thus, we can use Case A. Use the direct substitution method:

Test the state controllability:

$$|\mathbf{M}|=\begin{vmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \end{vmatrix}=\begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -3 & 7 \end{vmatrix}=-1$$

Thus, the system is completely state controllable.

$$\mathbf{K}=\begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}$$

$$|s\mathbf{I}-\mathbf{A}+\mathbf{BK}|=\left|\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix}-\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}+\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}\right|$$

$$|s\mathbf{I}-\mathbf{A}+\mathbf{BK}|=\begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ k_1 & 2+k_2 & s+3+k_3 \end{vmatrix}$$

$$|s\mathbf{I}-\mathbf{A}+\mathbf{BK}|=s^3+(3+k_3)s^2+(2+k_2)s+k_1=0$$

$$s^3+(3+k_3)s^2+(2+k_2)s+k_1=(s-\mu_1)(s-\mu_2)(s-\mu_3)$$

$$s^3+(3+k_3)s^2+(2+k_2)s+k_1=s^3+14s^2+56s+160$$

$$3+k_3=14,\quad 2+k_2=56,\quad k_1=160\Rightarrow \mathbf{K}=\begin{bmatrix} 160 & 54 & 11 \end{bmatrix}$$

Type 1 Servo Systems – Case A

The servo system can be represented as:

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}k_1 r$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -160 & -56 & -14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 160 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$u(\infty) = -\mathbf{K}\mathbf{x}(\infty) + k_1 r = 0$$

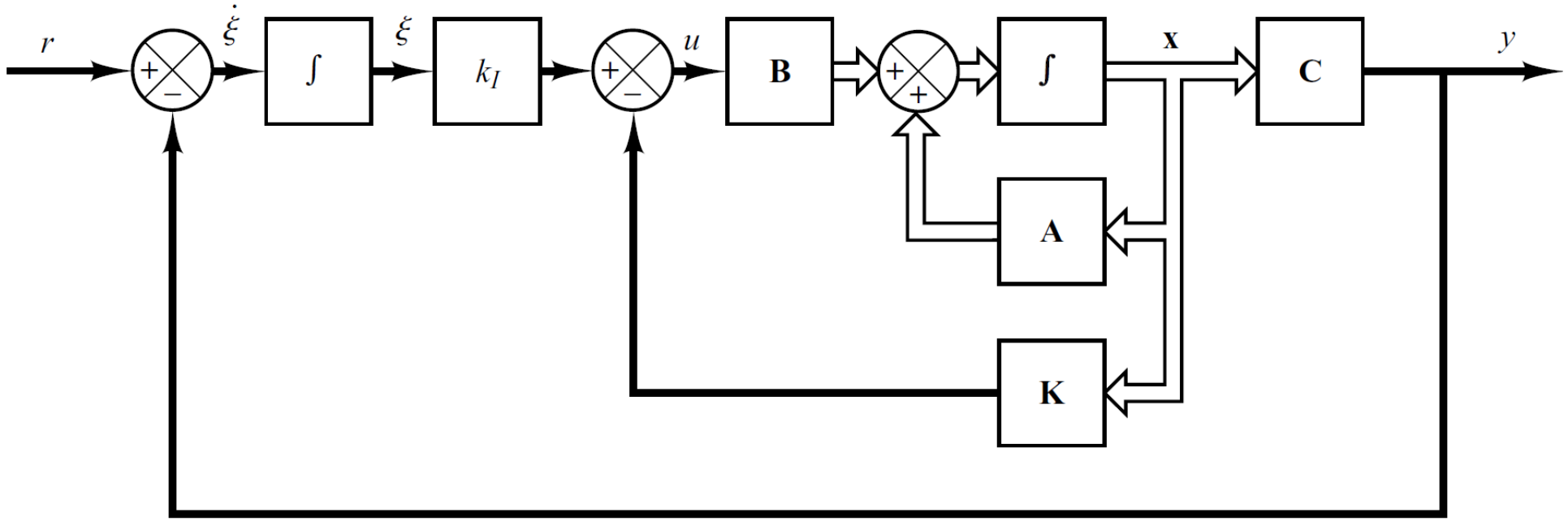
$$u(\infty) = -\begin{bmatrix} 160 & 54 & 11 \end{bmatrix} \begin{bmatrix} x_1(\infty) \\ x_2(\infty) \\ x_3(\infty) \end{bmatrix} + 160r$$

$$u(\infty) = -\begin{bmatrix} 160 & 54 & 11 \end{bmatrix} \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix} + 160r = 0$$

Type 1 Servo Systems – Case B

STRUCTURE

A Type 1 – Case B servo system when the plant does not have an integrator can be represented as follows:



Type 1 Servo Systems – Case B

RELATIONSHIPS

Consider the completely state controllable system:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$
$$y = \mathbf{C}\mathbf{x} + Du$$

The control signal u:

$$u = -\mathbf{K}\mathbf{x} + k_I\xi$$

where ξ is the output of the integrator, and:

$$\dot{\xi} = r - y = r - \mathbf{C}\mathbf{x}$$

The system dynamics can be represented by:

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\xi} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{-C} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \xi \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} u + \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} r$$

Type 1 Servo Systems – Case B

RELATIONSHIPS

The output equation is:

$$y = \begin{bmatrix} \mathbf{C} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \xi \end{bmatrix}$$

The error dynamics can be given as:

$$\dot{\mathbf{e}} = \hat{\mathbf{A}}\mathbf{e} + \hat{\mathbf{B}}u_e$$

where:

$$\hat{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{-C} & 0 \end{bmatrix}, \quad \hat{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix}$$

Thus:

$$u_e = -\hat{\mathbf{K}}\mathbf{e}$$

Type 1 Servo Systems – Case B

RELATIONSHIPS

where:

$$\hat{\mathbf{K}} = \begin{bmatrix} \mathbf{K} & -k_I \end{bmatrix}$$

The error dynamics become:

$$\dot{\mathbf{e}} = (\hat{\mathbf{A}} - \hat{\mathbf{B}}\hat{\mathbf{K}})\mathbf{e}$$

The system is fully state controllable if the matrix $\mathbf{P}$ is a full rank:

$$\mathbf{P} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{-C} & 0 \end{bmatrix}$$

The rest of the design process can be carried out using Ackermann's Formula, for instance.

Type 1 Servo Systems – Case B

EXAMPLE 2

Design a type 1 servo system for the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Assume that the desired closed-loop poles are at:

$$s = -5 + j5, \quad s = -5 - j5, \quad s = -10$$

Then, investigate the response of the system using MATLAB using a step input.

Type 1 Servo Systems – Case B

Find the poles of the system:

$$|s\mathbf{I}-\mathbf{A}|=\begin{vmatrix} s+3 & 2 \\ -1 & s \end{vmatrix}=s^2+3s+2=(s+1)(s+2)=0$$

The system does not include an integrator. Thus, we can use Case B.

Test the state controllability:

$$|\mathbf{M}|=|\mathbf{B}\quad \mathbf{AB}|=\begin{vmatrix} 1 & -3 \\ 0 & 1 \end{vmatrix}=1$$

Thus, the system is completely state controllable.

$$\hat{\mathbf{A}}=\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{-C} & 0 \end{bmatrix}=\begin{bmatrix} -3 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix}$$

$$\hat{\mathbf{B}}=\begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},\quad \hat{\mathbf{C}}=\begin{bmatrix} \mathbf{C} & 0 \end{bmatrix}=\begin{bmatrix} 0 & 2 & 0 \end{bmatrix}$$

$$\mathbf{P}=\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{-C} & 0 \end{bmatrix}=\begin{bmatrix} -3 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix}$$

P is a full rank. Thus, the system is completely state controllable.

$$\hat{\mathbf{K}}=\begin{bmatrix} \mathbf{K} & -k_I \end{bmatrix}$$

$$(s-\mu_1)(s-\mu_2)(s-\mu_3)=s^3+20s^2+150s+500$$

$$\alpha_1=20,\quad \alpha_2=150,\quad \alpha_3=500$$

Type 1 Servo Systems – Case B

Solve using Ackermann's Formula:

$$\phi(\hat{\mathbf{A}}) = \hat{\mathbf{A}}^3 + \alpha_1\hat{\mathbf{A}}^2 + \alpha_2\hat{\mathbf{A}} + \alpha_3\mathbf{I} = \hat{\mathbf{A}}^3 + 20\hat{\mathbf{A}}^2 + 150\hat{\mathbf{A}} + 500\mathbf{I}$$

$$\phi(\hat{\mathbf{A}}) = \begin{bmatrix} -3 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix}^3 + 20\begin{bmatrix} -3 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix}^2 + 150\begin{bmatrix} -3 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix} + 500\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 175 & -194 & 0 \\ 97 & 466 & 0 \\ -34 & -296 & 500 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{B}} & \hat{\mathbf{A}}\hat{\mathbf{B}} & \hat{\mathbf{A}}^2\hat{\mathbf{B}} \end{bmatrix} = \begin{bmatrix} 1 & -3 & 7 \\ 0 & 1 & -3 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\hat{\mathbf{K}} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \hat{\mathbf{B}} & \hat{\mathbf{A}}\hat{\mathbf{B}} & \hat{\mathbf{A}}^2\hat{\mathbf{B}} \end{bmatrix}^{-1}\phi(\hat{\mathbf{A}})$$

$$\hat{\mathbf{K}} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -1.5 \\ 0 & 0 & -0.5 \end{bmatrix}\begin{bmatrix} 175 & -194 & 0 \\ 97 & 466 & 0 \\ -34 & -296 & 500 \end{bmatrix}$$

$$\hat{\mathbf{K}} = \begin{bmatrix} 17 & 148 & -250 \end{bmatrix}$$

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State Observers - 1

Next Week …

DO YOU HAVE ANY QUESTIONS ?